

Divyesh Patil

Full Stack Engineer | Web & Mobile | AI Integration

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 2.5+ years of experience building, scaling, and deploying end-to-end web and mobile applications across diverse domains. Specialized in backend system design, cloud infrastructure (AWS), RESTful and real-time API development, and AI/LLM integration within production-grade systems. Experienced in the complete development lifecycle — from architecture design and database modeling to containerized deployment and third-party service orchestration. Delivered production systems across healthcare, fintech, e-commerce, and entertainment verticals with measurable performance and scalability outcomes.

TECHNICAL SKILLS

Backend: FastAPI, NestJS, Laravel (PHP), Django, Python, RESTful APIs, WebSocket, SSE Streaming, Redis, Supabase Edge Functions

Frontend: React, React Native (Expo), Redux, Angular, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, Material-UI, SvelteKit, Three.js, BPMN.js

Database & Cloud: MySQL, PostgreSQL, Supabase, Firebase, AWS (S3, EC2, RDS), Docker, Docker Compose, Git & GitHub, CI/CD

AI & Integrations: OpenAI API, OpenAI Whisper, Deepgram, LLM Integration, Anthropic Claude, Ollama, PostHog, Loops API, Telegram Bot API, SendGrid, TMDB API

Security & Payments: JWT Authentication, OAuth2 (Google & GitHub), Razorpay, Apple IAP, Google Play Billing, RevenueCat, Coinremitter, Webhook Architecture, Presigned URLs

Mobile & DevOps: React Native, Expo, App Store & Google Play Deployment, Docker Compose, Multi-tenant Architecture, Pytest, ERD Design

PROFESSIONAL EXPERIENCE

Software Engineer (Full Stack) — [fxis.ai](#)

Feb 2024 – Present

Intern: Feb 2024 – Jul 2024 | Full-Time: Aug 2024 – Present | Total: 2.5 Years

Biomni & Open WebUI — Agentic AI Medical Platform • SvelteKit • FastAPI • Docker Compose • Ollama • Anthropic Claude • Pytest

- **Platform Architecture:** Designed, containerized, and deployed a multi-tenant AI medical platform using SvelteKit, FastAPI, and a Biomni streaming server running local Ollama models (qwen3:14b, llama3.2) and Anthropic Claude, orchestrated via Docker Compose on a shared network.
- **RBAC & Multi-Tenancy:** Engineered a granular 4-tier Role-Based Access Control system (Admin, Organization, Invitee, Self) featuring organization-scoped API key management, key-preference switching, a 10-character referral code invite system, and rolling 7-day quota enforcement (50 credits/week).
- **Secure Registration Pipeline:** Built SMTP OTP verification and an AI-driven institutional questionnaire for automated biomedical organization approval, with SSE streaming for real-time AI completions.
- **Test Coverage:** Authored 74 Pytest integration tests covering end-to-end RBAC and access-control scenarios, ensuring stability across all permission boundaries.
- **Open-Source Contribution:** Contributed to Stanford SNAP's Biomni project by refactoring `query_reactome()` to support 2025 Reactome API `/search/query` endpoints, eliminating legacy 404 pathway/gene lookup errors.

ThroughLine — Mental Health Companion App • React Native • Expo • FastAPI • Supabase • RevenueCat • PostHog

- **Full-Stack Mobile Product:** Architected and shipped a production React Native (Expo) mental health app to the App Store, integrating a real-time streaming AI chat interface, native voice recording via Expo AV, and a structured therapy session planner.
- **Backend Engineering:** Built and optimized 3 core FastAPI endpoints — streaming LLM chat, Deepgram voice transcription, and AI-powered planner topic generation — supporting real-time mobile interactions at production scale.
- **Subscription & Monetization:** Integrated RevenueCat SDK and engineered FastAPI webhook handlers to process Apple IAP lifecycle states, enforce paywalls, and automate subscription conversion and lapse recovery flows.
- **Notification System:** Architected an automated push notification system using Supabase Edge Functions and Expo Push API with user-configurable scheduling, snooze/dismiss logging, and a client-side opt-in modal.
- **Analytics:** Implemented GDPR-compliant PostHog analytics with manual screen tracking and scoped touch autocapture, ensuring privacy compliance across all tracked user interactions.

Valunova Medical Sales System • Laravel • MySQL • AWS • ERD Design

- **Database Architecture:** Designed a normalized enterprise relational database managing a 3-tier distribution hierarchy (device owners → distributors → sales representatives) with comprehensive ERD documentation covering all business relationships.

- **Business Logic:** Implemented automated invoice generation, real-time inventory tracking, multi-level commission calculation, and granular sales reporting — eliminating manual reconciliation across the distribution chain.
- **Access Control:** Built role-based access control across all organizational tiers ensuring strict data isolation, audit logging, and compliance at the enterprise level.

Haveto Crypto Payment Gateway · Laravel · Coinremitter · Webhook · SendGrid

- **Payment Infrastructure Migration:** Architected and executed the migration from Copperx to Coinremitter, enabling Bitcoin, Ethereum, and Dogecoin support with zero service disruption during the transition window.
- **Real-Time Monitoring:** Built a webhook infrastructure for instant transaction state updates, integrated SendGrid for automated payment confirmation emails, and designed a UI supporting seamless dual-gateway operations during migration.

Astrology Chatbot App · React Native · Laravel · AWS EC2/RDS · Redis · Razorpay

- **Backend & Infrastructure:** Designed and scaled a Laravel (PHP) RESTful backend on AWS EC2/RDS with Redis caching, custom middleware, optimized DB migrations, and non-blocking background schedulers via Laravel Kernel for high-concurrency AI workloads.
- **Subscription Billing:** Architected a Razorpay-powered multi-tenant subscription system supporting multiple product tiers, webhook-driven payment lifecycle management, and automated account maintenance operations.
- **Admin Analytics:** Built a React dashboard with real-time AI usage metrics populated by server-side logging triggers on every model invocation, enabling data-driven product decisions.
- **Mobile Delivery:** Managed end-to-end publishing pipeline for both Google Play Store and Apple App Store, including versioning, compliance, and release management.

BPMN Diagram Generator · React · FastAPI · WebSocket · OpenAI Whisper · Docker

- **AI-Powered System:** Engineered a full-stack application converting text (200 words) and audio input (2-minute WebM) into valid BPMN 2.0 XML diagrams via OpenAI Whisper speech-to-text and LLM-driven structured XML generation, deployed as a multi-tenant Docker Compose application.
- **Real-Time UX:** Implemented WebSocket streaming for live processing feedback with tenant isolation and secure session management across concurrent users.

Face Shape Detection App · React · FastAPI · AWS S3/EC2/RDS

- **Full-Stack AI Application:** Architected a face shape classification platform with FastAPI backend and React frontend, featuring drag-and-drop upload, live camera capture, and AWS S3 presigned URL integration for secure personalized recommendation delivery.
- **Cloud Deployment:** Deployed and managed on AWS EC2 with RDS, supporting AI-generated recommendations, accuracy reporting, blog integration, and regulatory compliance pages in production.

Trip Management Mobile App · React Native · Google Play · App Store

- **Mobile Development:** Built a React Native application with a complex multi-screen navigation architecture; resolved critical back-stack bugs affecting user retention and published updates to Google Play Store and Apple App Store following platform compliance guidelines.

FX Company Website Revamp · JavaScript · HTML5 · CSS3 · WebP · Figma

- **Performance Engineering:** Led full website redesign achieving a 40% performance improvement through WebP image optimization, CSS sprite techniques, and JavaScript minification; delivered interactive map visualization with custom markers and Figma-to-production UI components.

Ski-Lift Status Bot — Japan · Python · Docker · Telegram Bot API · OpenAI

- **Intelligent Monitoring:** Built a Dockerized automated monitoring system scraping 5 Japanese ski resorts every 15–20 minutes, with change-detection algorithms and OpenAI-powered Telegram notifications delivering contextual lift status and snow condition summaries.

Solaris Finance Platform · Laravel · React · PHP · MySQL · Client Contract via FXIS.ai

- **Client Contract (3 months):** Worked on a live production Laravel + React financial platform through a contract engagement at FXIS.ai — diagnosed and resolved critical production bugs, delivered UI performance improvements, and led migration of legacy Laravel Blade-based views into modern React components while shipping feature enhancements with zero downtime.
- **Technical Scope:** Worked across the full stack — PHP/Laravel backend routes and controllers, React frontend components, and MySQL queries — requiring rapid onboarding into an existing production codebase and careful regression testing before every deployment.

FREELANCE

NE Vernacular AI — Multilingual AI Platform · React · TypeScript · FastAPI · Node.js · Sarvam AI · Gemini · HeyGen · Docker Compose

- **Architecture:** Designed and built a 3-service containerised multilingual AI platform using Docker Compose — React/TypeScript (Vite) frontend, Express/Node.js TypeScript proxy backend with JWT auth middleware, and a Python FastAPI AI service handling all AI operations.

- ▶ **Multi-Provider AI Pipeline:** Integrated Sarvam AI as the primary provider for translation (sarvam-translate:v1), TTS (Bulbul v3 with language proxy mapping for unsupported scripts), and ASR (Saaras v3 speech-to-text); used Google Gemini as an automatic fallback for rare NE Indian scripts (Khasi, Mizo, Tripuri, Meitei) — enabling coverage across Assamese, Bodo, Manipuri, Nepali, and 8+ more languages.
- ▶ **Dual ASR Input:** Implemented two speech-to-text modes — browser-native Web Speech API with real-time interim results and audio level visualisation (zero backend cost), and server-side ASR via file upload to Sarvam Saaras v3 for higher accuracy.
- ▶ **Video Generation:** Built HeyGen AI avatar video generation and Seedance/Fal AI scene video generation pipelines with async job polling; added a translation caching layer to eliminate redundant API calls and reduce operating costs.

Nose Shape Detection App · FastAPI · React · AI Classification

- ▶ Developed an AI-powered web application classifying 12 nose shape categories across 5 analytical parameters, with dual image input (drag-and-drop and live camera capture), a FastAPI backend, educational content, and regulatory compliance pages.

KEY PERSONAL PROJECTS

Cine AI — Movie Recommendation Platform · React · Redux · Firebase · OpenAI · TMDB API

- ▶ AI recommendation engine using TMDB API with trailer autoplay, Firebase authentication, Redux memoization, and OpenAI (4o-mini) for intelligent personalized movie recommendations with multi-language search and Open Graph social sharing.

Authentication System · JWT · OAuth2 · React

- ▶ **Comprehensive authentication framework:** email/password, OTP verification, password reset, dual OAuth (Google & GitHub), JWT token management with secure refresh logic, and full error handling across all authentication flows.

Custom Food Ordering Platform · Django · React · Redux · MySQL · Razorpay

- ▶ Full-stack e-commerce application with Django REST backend, React/Redux frontend, Razorpay gateway integration, and a role-based admin dashboard for order, user, inventory, and dish customization management.

EDUCATION

Bachelor of Engineering — Computer Engineering

Graduated: 2024

Gandhinagar Institute of Technology | CGPA: 8.02

CERTIFICATIONS & ACHIEVEMENTS

- ▶ Python Essential 1 & 2 — Cisco Networking Academy
- ▶ Python 101 for Data Science — IBM
- ▶ Deployed full-stack production systems on AWS (EC2, RDS, S3) across healthcare, fintech, and entertainment domains
- ▶ Published 4+ applications on Google Play Store and Apple App Store; managed complete store compliance lifecycle
- ▶ Achieved 40% website performance improvement through WebP, sprite, and minification optimization techniques
- ▶ Designed enterprise-scale normalized database architectures with comprehensive ERD documentation
- ▶ Contributed to Stanford SNAP open-source project (Biomni) — Reactome API production fix
- ▶ Built and integrated AI/LLM systems across 8+ production applications spanning chatbots, automation, and domain-specific platforms